

[DC] Data-Driven Expertise Assessment in XR: Analyzing Multimodal Sensor Data for Psychomotor and Cognitive Tasks

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ABSTRACT

In this doctoral research, a novel approach is developed for the real-time assessment of user expertise, focusing on skill-based and cognitive tasks within Extended Reality (XR) environments. This approach combines advanced user modeling with state-of-the-art sensors and includes an in-depth analysis of gaze behavior and physiological responses. A distinctive aspect of this research is the real-time analysis of multimodal sensor data, which provides deeper and more precise insights into user skills and cognitive abilities. The goal is to advance the field of expertise assessment by introducing a nuanced and dynamic perspective that is suitable for a variety of applications across different domains. This research aims to establish new standards in the assessment of user expertise, meeting the evolving needs of modern educational and professional settings.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / Augmented reality; Computing methodologies—Pattern recognition—Biometrics; Applied computing—Education—Interactive learning environments

1 INTRODUCTION

Advances in Extended Reality (XR) technology is reshaping our interaction and experience with reality [3], offering unprecedented capabilities for learning in both professional and educational contexts. This Ph.D. research addresses the fundamental problem associated with the real-time assessment of user expertise in skill-based and cognitive tasks, guided by XR systems. This research integrates advanced user modeling and sensor technologies within XR environments, offering a nuanced approach to address this challenging problem. Building on recent developments in information technology and artificial intelligence, coupled with enhanced computing power and high resolution displays, the proposed research aims to answer an overarching question: *Can we build accurate and scalable data-driven models for online and task-specific estimation of user expertise based on multimodal data streams captured by XR headsets and wearables?*

The proposed research is motivated by the need for a context-aware understanding of user activities and metacognitive states (e.g. affects, emotions, cognitive load [5]), particularly in industrial scenarios such as aviation, manufacturing, or healthcare, which involve complex psychomotor tasks. The vision is to incorporate the principles of multimodal machine learning for the effective processing and analysis of sensor data streams [1], [2], [4]. By developing a multifaceted and online expertise assessment tool, this research addresses a significant gap in the state-of-the-art in user modeling in XR, with particular emphasis on workplace-based learning use cases. Furthermore, this research will establish new benchmarks in the domain of expertise assessment by formulating new tasks and annotated datasets. This approach is designed to offer educators, trainers, and industry professionals a useful and scalable tool to

assess and improve skills in cognitive psychomotor tasks in various educational and industrial environments.

2 RESEARCH CONTRIBUTIONS

This research introduces several important advancements in the assessment of expertise within XR environments aimed at accelerating learning, characterized by the following key developments:

1. **Integration of Sensors for Real-Time Multimodal Machine Learning in XR:** Research stands out for the integration of various sensors to simultaneously analyze behavioral, physiological and cognitive data in real time within XR for expertise assessment. This approach provides a comprehensive and immediate understanding of user expertise utilizing the depth and breadth of multimodal data.
2. **Development of Individual User Profiles for Cognitive Task Performance:** A significant advancement is the adaptive creation of user profiles for cognitive task assessments within XR environments. The focus is on building a system that can accurately assess and estimate individual expertise levels, based on unique performance patterns and responses.
3. **Holistic Approach to Expertise Assessment:** The approach adopted in this investigation encompasses the evaluation of both psychomotor and cognitive skills, using advanced sensor data for the assessment of user expertise with respect to an array of criteria and metrics. This comprehensive approach aims to address existing gaps in data-driven expertise assessment, providing a more comprehensive and objective approach for the assessment of user skills and abilities.
4. **Application in XR Environments for Expertise Assessment:** The innovative use of XR technologies for expertise assessment makes a significant contribution to the field, providing an accurate online tool applicable to various professional and educational contexts.

Collectively, these contributions represent a significant advancement in expertise assessment technology, setting the stage for future innovations and applications in the field.

3 POTENTIAL USE CASES

This research will specifically explore two industrial use cases without loss of generality. The rationale here is to study different aspects or dimensions of expertise pertinent to these (and similar) use cases.

3.1 Use Case 1: Aviation Training—Flight Simulations

Context: Technical skill and decision-making mastery in pilot training. **Task:** Completing a simulated flight, including emergency responses. **Measurements:** (1) Technical proficiency (e.g., flight maneuver accuracy). (2) Crisis management (e.g., emergency response, stress management). (3) Physiological responses (e.g., heart rate variability). (4) Attention and reaction times (e.g., situational awareness). **Outcome:** Assessment of pilot capabilities for real-world readiness.

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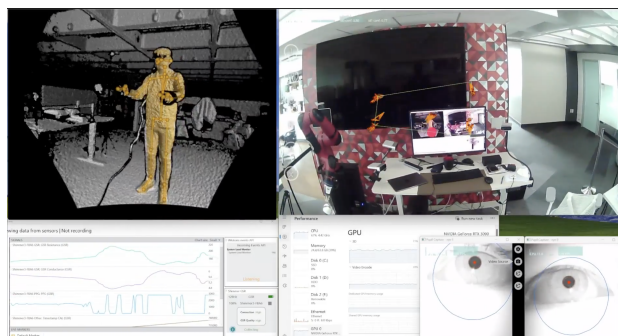


Figure 1: Prototype XR system: Integration of advanced sensors and user modeling tools within the XR platform.

3.2 Use Case 2: Industrial Training—Machine Operation

Context: Precision training for complex machinery operation in industrial settings. **Task:** Operating virtual industrial machinery for tasks like assembly. **Measurements:** (1) Operational proficiency (e.g., machine operation skills). (2) Problem-solving abilities (e.g., diagnosing, fixing issues). (3) Cognitive and physical load (e.g., stress and fatigue monitoring). (4) Spatial awareness (e.g., machinery interaction, workspace awareness). **Outcome:** Proficiency insights for training and workplace safety.

4 CURRENT PROGRESS AND PRELIMINARY FINDINGS

The development of a prototype XR system marks a significant milestone in this research. Engineered to collect complex real-time data, the system utilizes advanced technologies such as the Varjo XR-3, the Pupil Labs HoloLens add-on, and the Microsoft Azure Kinect DK camera. It is further enhanced by Unity Engine's virtual GameObjects to provide an interactive and comprehensive platform to assess expertise (see Figure 1). This prototype pipeline is designed to capture a wide range of metrics, including gaze behavior metrics such as pupil size, gaze velocity, fixations, and saccades, along with physiological measures such as electrocardiogram (ECG), photoplethysmogram (PPG), and galvanic skin response (GSR) (see Figure 2). The integration of these multimodal data collection technologies is crucial for the creation of individual user profiles that encompass behavioral patterns and cognitive load, vital for understanding individual performance and learning trajectories.

In the initial phase, the main focus has been fine-tuning and validating the data collection process to ensure that the data gathered are accurate and reliable. This involves integrating advanced user modeling with cutting-edge sensors. Early results are encouraging, showing the prototype's capability in assessing expertise accurately in cognitive psychomotor tasks. Initial findings align with studies on AR technologies, particularly eye tracking and gaze behavior analysis in AR-guided tasks [6]. Key observations include distinct gaze behaviors between novices and experts and connections between pupil size and fixation/saccade patterns, crucial for AR training. Future development aims to refine the precision of the system and user experience, with the goal of creating a versatile and reliable expertise assessment tool for diverse applications.

5 CONCLUSIONS AND FUTURE DIRECTIONS

This research is envisioned to significantly advance the field of expertise assessment by harnessing XR technology coupled with multimodal machine learning. Through the integration of user modeling and advanced sensor technologies within XR environments, this research will contribute novel approaches to objectively evaluating skills and cognitive abilities in real time. This will be particularly impactful in complex, psychomotor tasks in aviation, manufacturing,

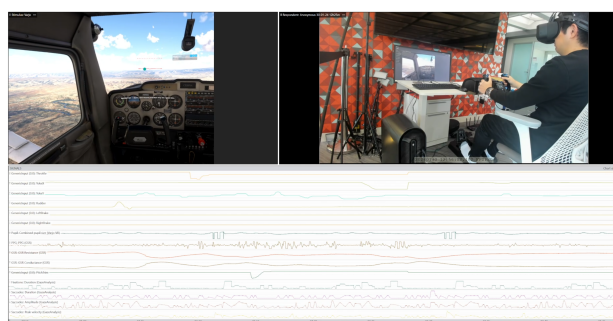


Figure 2: XR training setup: Integrating Varjo XR-3 eye tracking, Shimmer3 GSR+ physiological sensors, and controller data with MS Flight Simulator for real-time assessment of expertise.

healthcare, and other industry sectors. The overarching goal is to create accurate and nuanced user profiles, which, in turn, would allow XR to offer personalized learning experiences. Current research has demonstrated the effectiveness of combining user modeling with state-of-the-art sensors to meet the evolving needs of modern educational and professional settings. Future directions include the development of robust frameworks for integrating and processing multimodal sensor data, which will enhance the system's efficiency in handling large datasets and enable more effective real-time analysis. Additionally, a major focus will be placed on refining user modeling techniques to more accurately assess individual expertise levels in various scenarios. The vision is to establish a foundational XR-based system for expertise assessment, tailored for adaptability and effectiveness in diverse settings. The upcoming Doctoral Consortium is expected to provide crucial information for further refinement of the proposed framework and system, particularly in improving its ability to accurately assess expertise in complex psychomotor tasks.

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